AQUAFAB INDUSTRIES

Fish Tracking

Following an animal through a video, and measuring how it swam.

For Aquafab Workbench 0.1.0

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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1 Fish tracking

Fish tracking finds the animal in each frame of a video by comparing it with an empty-tank background, joins the detections into a track, and measures position, swim speed and distance travelled. It reads a recorded clip or a live camera and does not drive any machine, so it works on video filmed anywhere. It needs a license.

Where to find it:

- **Experiments → Fish Tracking** on a machine: runs are saved to that machine's Trial History.
- **Video Analysis** on the Launcher: no machine needed. Runs are *not* saved to Trial History; keep them with **Export Track…**.

> Screenshots use a synthetic video made for this guide. The speeds in them are not measurements.

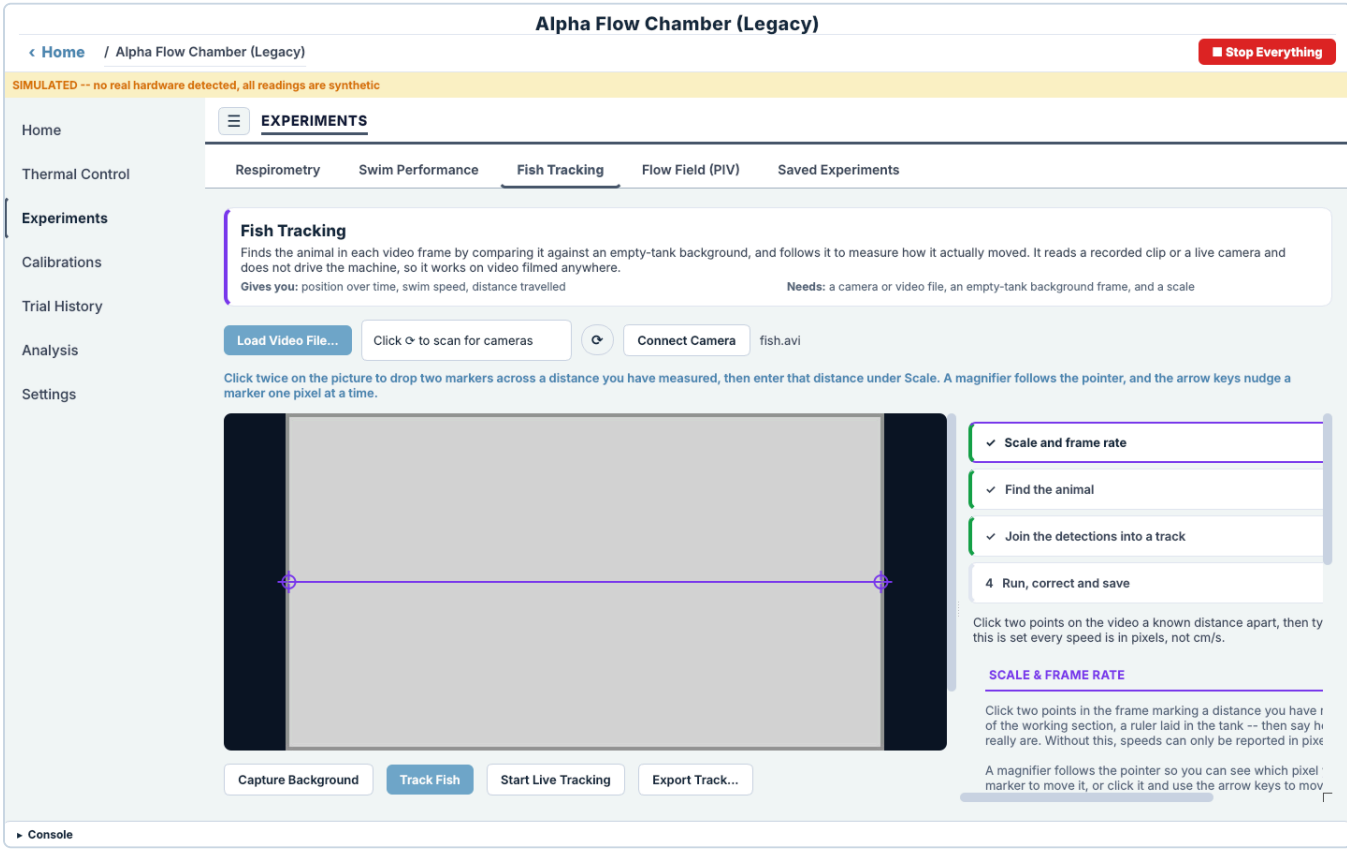
1.1 Filming for tracking

- **Fix the camera.** If it moves after the background is captured, everything that moved looks like an animal.
- **Light evenly**, ideally from below or behind, for a dark animal on a bright, even field.
- **Record a few seconds with no animal in view**, or film the empty tank first. That is the background.
- **Put a known distance in view**: a ruler, or the working section's width.

1.2 1. Load the video

Load Video File… (`.avi`, `.mp4`, `.mov`, `.mkv`, `.m4v`), or choose a camera and **Connect Camera** (⟳ rescans).

1.3 2. Scale and frame rate



Setting the scale.

1. Click one end of the known distance, then the other. A magnifier shows the exact pixel. Drag a marker, or select it and use the arrow keys (Shift for ten pixels).
2. Type **Distance between markers** (cm) and press **Apply scale from markers**.
3. Leave **Frame rate** at 0 unless the camera reports it wrongly.

Without a scale, speeds are in pixels.

1.4 3. Find the animal

1. Step to a frame with no animal in view and press **Capture Background**.
2. Adjust **Detection**:

Field	What it does
Threshold	How different from the background a pixel must be to count as animal. Raise it first if noise is detected.
Min blob area (px²)	Ignore specks smaller than this.
Limit max blob area	Ignore large shapes, such as shadows.
Blur kernel size	Smooths sensor noise. 1 = off.

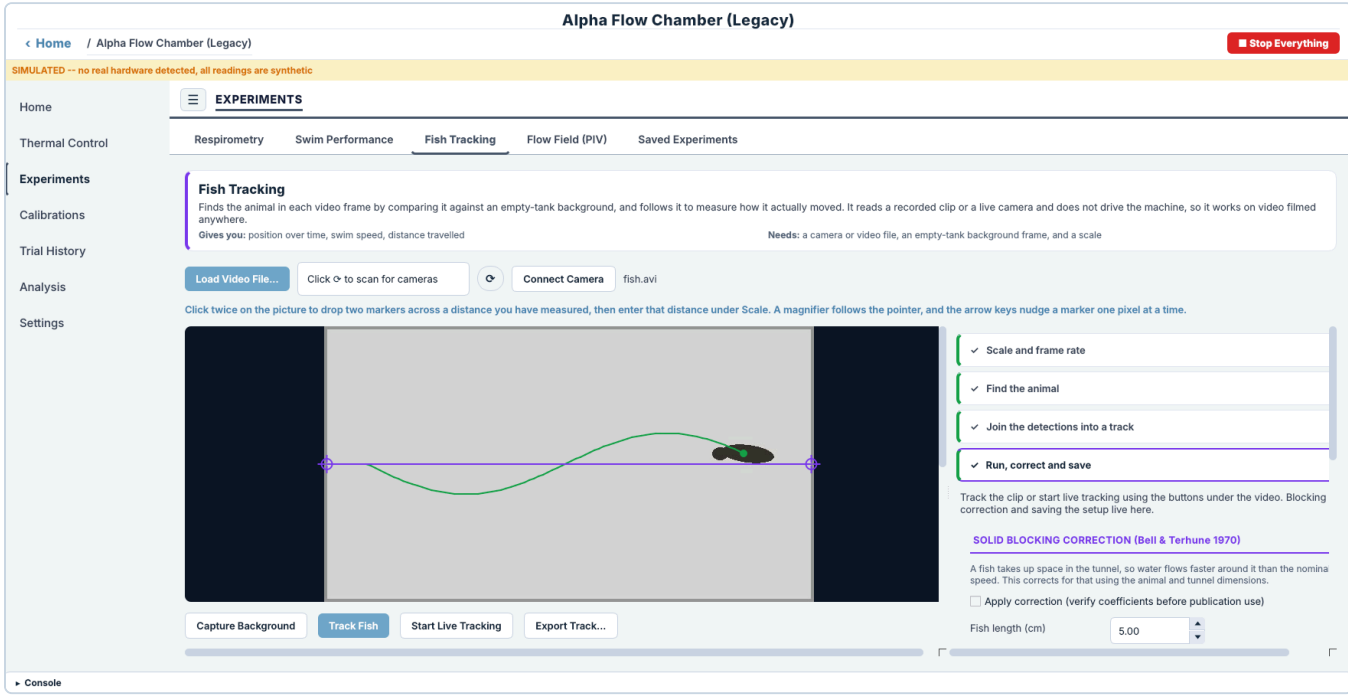
Field	What it does
Morphology kernel size	Makes the animal one solid blob. 1 = off.
Frame stride	Analyse every Nth frame, for speed on long clips.

1.5 4. Join the detections into a track

Off by default, the largest detection in each frame is the animal. Tick **Multiple fish** to follow every animal, and set **Search range** to the furthest an animal moves between frames: too large and tracks swap.

1.6 5. Run and save

Button	What it does
Track Fish	Tracks the whole clip, draws the track and the speed chart, and gives mean and maximum speed. On a machine, saves the run to Trial History.
Start Live Tracking	Plots swim speed continuously from the camera.
Export Track…	CSV: frame, time, x, y and speed for each frame (and corrected speed, below).



A tracked clip and its swim speed.

Check the drawn track. A track that jumps across the tank and back has picked up something that is not the animal: raise the threshold or minimum area, or capture a fresh background.

1.6.1 Solid blocking correction

In a swim tunnel, the animal blocks part of the working section and the water around it flows faster than the nominal speed. Tick **Apply correction** and enter **Fish length**, **thickness** and **cross-section**, and the **Tunnel cross-section** (from the machine's Settings where measured). The **Blockage exponent** is 1.0 or 1.5 depending on the source; verify against Bell & Terhune (1970) before publishing.

1.6.2 Keep the setup

Name, **Treatment** and **Save experiment** keep the detection, linking and blocking settings, so a setup that took a few clips to settle can be reused on any machine.

1.7 Finding past runs

Analysis → Fish tracking lists every saved tracking run. **Open in Fish Tracking** reloads the clip and its settings.